

Assignment

COI-AT-1

Short Answer test.

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Course : Fundamentals
of Data Science.

Code : DSA0405

Short Answer Questions:

1. What is Data science? Explain its primary objective.

Data science is the process of collecting, analyzing and interpreting data using scientific methods, statistics and machine learning to extract useful insights.

Primary objective:

To convert raw data into meaningful information for better decision-making.

2. Why is data science important in today's digital world? Mention any four reasons.

1. Helps in better decision-making.
2. Predicts future trends and outcomes.
3. Improves business efficiency
4. Enables personalized customer experiences.

3. List any four benefits of Data Science in business and society.

1. Improves business performance.
2. Reduces operational costs.
3. Enhances healthcare services.
4. Supports fraud detection and security.

4. Mention four real-world applications of data science.

1. Healthcare - Disease prediction.
2. Banking - Fraud detection.
3. E-commerce - Product recommendations.
4. Transportation - Traffic and route optimization.

5. What are the different facets of data? Briefly explain each.

1. Structured data - organized in rows and columns. (Eg: database tables)

2. Semi-structured data - Partially organized with tags. (eg: XML, JSON)

3. Unstructured Data - No fixed format. (eg: images, videos, emails).

6. Differentiate between structured, semi-structured and unstructured data with suitable examples.

Type	Description	Examples
structured	organized in tables	student database
Semi-structured	Contains tags/labels	XML, JSON
unstructured	No predefined format	Images, Videos.

7. What is the big data? State its key characteristics (5Vs).

Big data refers to extremely large and complex datasets that cannot be processed using traditional methods.

5Vs:

* Volume. * Velocity. * Variety. * Veracity
* Value.

8. Explain the Big data ecosystem. Name any four technologies used in it.

The Big data ecosystem consists of tools used for collecting, storing, processing, and analyzing large datasets.

Technologies

* Hadoop. * Spark. * Hive * HBase.

9. what are the major phases of the Data science process?

1. Data collection.
2. Data preparation.
3. Data Analysis.
4. Model Building.
5. Model Evaluation.
6. Deployment.

10. what is data retrieval? Mention two common sources from which data can be retrieved.

Data retrieval is the process of obtaining data from storage or databases.

Sources:

1. databases.
2. Web APIs

11. Why is data collection considered an important step in the data science process.

Data collection is important because it provides accurate and relevant data for analysis, improving the quality and reliability of results.

12. What is data cleaning? List any four common data-cleaning techniques.

Data cleaning is the process of identifying and correcting inaccurate or incomplete data.

Techniques:

1. Remove duplicate records.
2. Handle missing values.
3. Correct inconsistent data.
4. Remove outliers/errors.

13. What is data integration? why it is necessary in data science?

Data integration is the process of combining data from multiple sources into a single dataset.

Need:

- * Provides a complete view of data.
- * Improves analysis accuracy.
- * Eliminates redundancy.
- * Supports better decision-making.

14. Explain data transformation with suitable examples.

Data transformation is the process of converting data into a suitable format for analysis.

Examples:

- * Converting text labels into numbers (1/0).
- * Normalizing values between 0 and 1.
- * Changing date format from DD/MM/YYYY to YYYY-MM-DD.

15. What is data analysis? Mention any four techniques used for data analysis.

Data analysis is the process of examining data to discover useful information and support decision-making.

Techniques:

1. Descriptive Analysis.
2. Diagnostic Analysis.
3. Predictive Analysis.
4. Prescriptive Analysis.